Strong Optical Disorder Erases Images but Preserves Local Change Observability

[AUTHORS][1]
[1] *[AFFILIATION]*
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Strong disorder is expected to erase spatial information. We show that reconstruction rank and local change observability are distinct invariants. After projecting out nuisances, the lowest order at which a change perturbs the statistics (its jet order) fixes the detection time. In a scrambled channel the scene Fisher matrix collapses to rank one, yet every nonzero local change stays visible: generic directions need $T \sim \varepsilon^{-2}$, orthogonal ones are curvature-rescued at $T \sim \varepsilon^{-4}$. Exact divergences, Monte Carlo, sequential detection, and a sealed simulated single-pixel optical test validate the result.

THE DISORDER PARADOX

A strongly scattering medium destroys the map between an object and its image. In the fully developed speckle limit the mean field carries only the object's total flux, and every attempt to invert the scrambled transmission fails once the angular memory effect is exhausted [1, 4]. The received wisdom is therefore that strong disorder erases spatial information outright. We show that this conflates two logically independent quantities. *Reconstruction rank*—how many scene coordinates a record can resolve—can collapse to one, while *local change observability*—whether an infinitesimal scene change alters the measured statistical law at all—remains complete. The physical statement of the Letter is that strong optical disorder can erase coordinates without erasing the observability of local change, and that a single invariant, the order of the first nonzero nuisance-profiled statistical jet, fixes the price of observing each direction.

The claim is not "detection is easier than imaging." It is a classification of physical observability by contact order that survives reparameterization and nuisance profiling, applies to any dominated statistical experiment, and is realized at its most extreme in complete optical scrambling. We prove a Rank–Jet Separation theorem, demonstrate its exact directional classes, and falsify it at its own boundaries in a sealed, fully simulated single-pixel optical model.

QUOTIENT INFORMATION JETS

Let one independent medium realization ("bank") generate a record Y with law $P_{x,\eta}$, where x is the static scene and η collects every admitted nuisance: the medium law, throughput, temporal, and detector degrees of freedom. Along a physical scene path $x_\varepsilon = x + \varepsilon\delta$ the relevant question is not whether a chosen observable moves, but whether the perturbed law leaves the nuisance orbit $N_x = \{P_{x,\eta'}\}$. We therefore profile every nuisance and work with the profiled Chernoff distance $C_*(\varepsilon;\delta) = \inf_{\eta'} C(P_{x+\varepsilon\delta,\eta}, P_{x,\eta'})$, whose infimum makes it invariant to nuisance coordinates, gain gauges, and changes of medium basis. If $C_*(\varepsilon;\delta) = c_m(\delta)\,\varepsilon^{2m} + o(\varepsilon^{2m})$ with $c_m > 0$, then

$$T_{\text{req}} \asymp \varepsilon^{-2m}, \tag{1}$$

and we call $m = m(\delta)$ the *quotient information-jet order*. For T independent-equivalent banks Chernoff's theorem gives $-T^{-1}\log P_e^*(T,\varepsilon) \to C_*(\varepsilon;\delta)$, so the fixed-error sample complexity obeys Eq. (1); the same profiled Kullback–Leibler expansion drives Lorden/CUSUM quickest detection [9], giving a mean delay $\propto \varepsilon^{-2m}$. The quotient jet is a stochastic observability order, the statistical counterpart of successive-derivative observability rank conditions [10]. The efficient profiled score attains the fluctuation–response bound [8]. The exponent is a property of the physical experiment after nuisance quotienting: it assumes no fixed trace, no isotropy, and no chosen estimator. Physical covariation changes the coefficient c_m; it cannot change the definition of m. This is the object that survives the spectral covariations which defeat any fixed-resource inequality.

RANK–JET SEPARATION

Suppose the marginal record law depends on the scene through r smooth functionals $q(x) = (q_1, \ldots, q_r)$, $P_x = P_{q(x),\eta}$, with every nuisance profiled out. Writing J_q for the efficient Fisher matrix in functional space, the scene Fisher matrix factors as

$$I_x = Dq_x^\top J_q\, Dq_x, \qquad \operatorname{rank}(I_x) \le r. \tag{2}$$

Thus $r = 1$ permits at most rank-one reconstruction information, regardless of how many detector samples are acquired. Reconstruction rank is bounded by the number of scene functionals the law exposes; it says nothing about the contact order of any one direction.

The strongest realization is the fully scrambled covariance channel. There a single nonnegative scalar $Q(x) = x^\top G x$ (a grain-weighted scene energy, $G \succ 0$) enters the record covariance, so

$$\Sigma(x) = R + Q(x)\,H, \qquad \Delta Q = 2\,x^\top G\delta + \delta^\top G\delta. \tag{3}$$

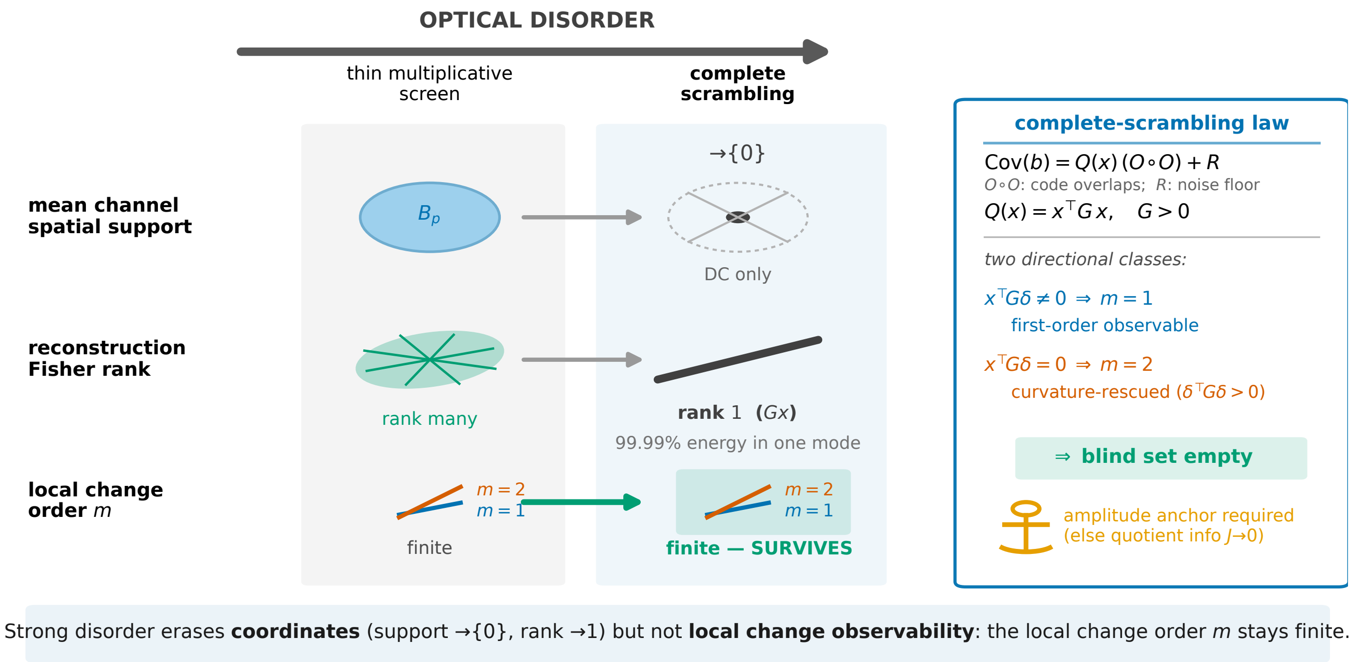

FIG. 1. **Rank collapses; jets survive.** As optical disorder grows from a thin multiplicative screen to complete scrambling, the mean-channel spatial support contracts $B_p \to \{0\}$ (DC only) and the reconstruction Fisher rank collapses from many modes to one (99.99% of the scene-dependent covariance energy along $O \circ O$), while the quotient jet order stays finite ($m = 1$ or 2) and the local blind set stays empty. At the scrambling endpoint the record covariance factors as $\mathrm{Cov}(b) = Q(x)\,(O \circ O) + R$ with $Q = x^\top G x$, and the two directional classes $u = 2x^\top G\delta$, $v = 2\delta^\top G\delta$ set the contact order. The amplitude-anchor condition (icon) is a prerequisite, not a caveat: without it the quotient information vanishes.

The quotient expansion of Q has first variation $u(\delta) = 2x^\top G\delta$ and curvature $v(\delta) = 2\delta^\top G\delta > 0$ for every $\delta \neq 0$. Because a regular reduced law inherits the contact order of Q—the reduced law is analytic in Q (End Matter A)—every direction falls into exactly one of two local classes,

$$\begin{aligned} x^\top G\delta \neq 0:\ & m = 1, \quad T_{\rm req} \sim \varepsilon^{-2}; \\ x^\top G\delta = 0:\ & m = 2, \quad T_{\rm req} \sim \varepsilon^{-4}. \end{aligned} \tag{4}$$

Hence *the reconstruction Fisher matrix is rank one, yet the local blind set is empty on the aperture where $G \succ 0$ (grain band covering the scene band)*: the first-order-orthogonal directions are not blind but curvature-rescued, since $\delta^\top G\delta > 0$. This is the Rank–Jet Separation theorem, proved in End Matter A.

Three boundaries fix its exact domain and make it stronger, not weaker. The statement is *local*: a signed finite change can return to the same Q ellipsoid at $\varepsilon_* = -2x^\top G\delta/\delta^\top G\delta$, where distinguishability vanishes. An unanchored covariance-amplitude nuisance renders the scene and amplitude derivatives collinear and collapses the quotient information to zero, so a wavelength, lag-shape, polarization, or finite calibration *anchor* is required. Global sign-definiteness holds only on the monotone cone $x, \delta \geq 0$ with an entrywise-nonnegative kernel; it is not claimed for arbitrary signed changes. The premise $G \succ 0$ requires the grain modulus-squared band $\hat{G} = |\widehat{C}|^2$ (supported on $|k| \leq 2k_{\rm grain}$) to cover the scene band; a change supported wholly beyond $2k_{\rm grain}$ gives $u = v = 0$ and is genuinely blind (Supplement, equality case (iii)).

SCRAMBLING INVERSION

Complete optical scrambling supplies the extreme endpoint. A band-limited modulator imprints a code p_i (Fourier support B_p) on the illumination; a dynamic strongly scattering medium with fully developed speckle then produces a zero-mean circular-complex-Gaussian field, and a single-pixel bucket sums the scene-weighted intensity. The mean speckle envelope is flat, so the first-moment channel measures only the scene DC term,

$$\mathbb{E}[b_i] = C(0)\, O_{ii}\, \hat{x}(0), \tag{5}$$

where O_{ii} is the code weight and $\hat{x}(0)$ the scene DC component; its spatial support is $B_{\rm mean} = \{0\}$: every non-DC change, in-band or beyond-band, is exactly invisible to the mean. Numerically the mean response to a beyond-band, DC-free change has relative magnitude 6.6×10^{-19}.

The intensity covariance, by the Siegert relation, factors into a known code part and the single scene scalar, $\mathrm{Cov}(b_i, b_j) = O_{ij}^2\, Q(x)$ with $Q(x) = x^\top G x$ and G the modulus-squared grain kernel. The support of G is enormous—the wavelength-scale grain pierces the modulator band—but the scene enters through the one scalar Q, so the reconstruction Fisher information is rank one and 99.99% of the scene-dependent covariance energy lies along the rank-one structure $O \circ O$. Beyond-band imaging dies with the memory effect [5, 6]. Yet by Eq. (3) any beyond-band change lifts Q by $2x^\top G\delta + \delta^\top G\delta$, and the quadratic floor is strictly positive for every nonzero δ: there is no blind direction for detection within the informative aperture. The mean wall rejects DC and flux changes; the covariance curvature supplies detection (Fig. 1).

DECISIVE TESTS

We test the theorem with the exact scrambling generator and no reconstruction, on a 32×32 grid ($N = 1024$, code band $P_B = 3$, grain band $K = 13$, $M = 36$ codes, 10^4 photons per bucket). Directions are constructed to be generic, exactly G-orthogonal, or mixed. Sweeping the amplitude over a decade and forming exact Gaussian statistical (KL and Chernoff) divergences, Monte Carlo log-likelihood-ratio error, and CUSUM delay yields the frozen result: exact KL orders were 2.038 and 4.000, Monte Carlo response exponents were 0.95 and 2.05, and CUSUM delay slopes were -2.16 and -3.92; the predicted crossover, nuisance collapse, anchor restoration, iso-Q cancellation, and monotone-cone corollary all passed their frozen tests (Fig. 2). Under a free medium-amplitude nuisance the efficient information for Q collapses to numerical zero and stays zero across proportional lags; a finite amplitude anchor restores it. At the signed iso-Q crossing the classifier returns to chance (AUC $\approx$ 0.48) while remaining visible on either side, confirming that the completeness is local. Over 400 in-aperture directions the blind fraction is exactly zero. All 17 predeclared checks passed and none of the four kill conditions fired.

SEALED OPTICAL INSTANTIATION

A preregistered simulated thin-screen study of the single-pixel bucket record converts the theorem into a power-certified sentinel. Under the sealed protocol (End Matter C), 2% beyond-band changes were predicted detectable in 77/81 cells (medium-parameter grid points); the best cell required 453 banks and achieved a Monte Carlo power lower bound of 0.990; a repeated code bank retained a latency advantage of 459 versus 1013 banks over fresh covariance codes; and online scoring consumed 0.16% of each bank interval. This is a power-certified beyond-band alarm with measured attribution leakage, not a fully specific sentinel: at the frozen 1% threshold the four-class balanced accuracy was 0.916 and in-band false alarms were 0.020, while medium-amplitude and medium-timescale events produced false-alarm rates of 0.096 and 0.084 (Fig. 3). For robustness, all frozen performance and calibration bars passed in the primary mismatch domain of aperture-preserving medium-basis rotations up to 10% (FA = 0.032, latency inflation 19%). At 20% rotation and a spectral-slope mismatch of -1, non-target false alarms rose to 0.052 and 0.076; these points define the measured robustness boundary, and no broader law-mismatch claim is made.

OUTLOOK

Identifiability dimension can vanish while local hypothesis distinguishability remains complete at a higher contact order. This invites classifying a statistical sensing experiment by three independent objects—its *support* (where the law can respond), its *rank* (how many scene coordinates it reconstructs), and its *jet* (the contact order at which a chosen direction becomes observable). Strong disorder can enlarge support, decrease rank, and leave jet order finite; no single scalar information budget captures this Disorder Observability Triangle, whose first physical realization is Fig. 1. A clean positive corollary bounds the honest scope: on the monotone cone ($x \geq 0$, $\delta \geq 0$, G entrywise nonnegative and $G \succ 0$), $Q(x+\delta) - Q(x) > 0$ for every $\delta \neq 0$, so local completeness becomes global for physically monotone addition processes such as deposition, coating growth, and damage accumulation. The result separates reconstruction dimension from observability order and transfers beyond optics to disordered and medium-change sensing [2, 7], sequential change detection [9], singular statistical models where finite-sample spectral emergence sets a second threshold [11], and quantum-inspired superresolution [3].

[FUNDING/ACKNOWLEDGMENTS]

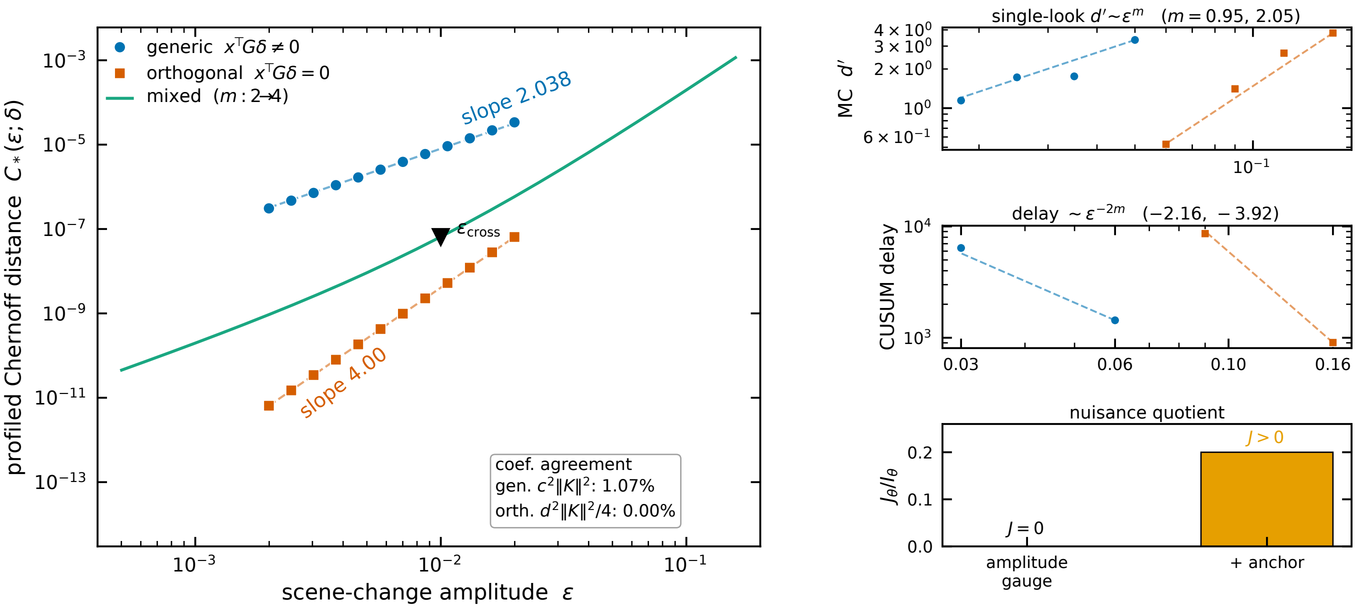

FIG. 2. **The exponent hierarchy.** Profiled Chernoff distance versus scene-change amplitude for a generic direction ($x^{\top}G\delta \neq 0$, $m = 1$) and a first-order-orthogonal direction ($x^{\top}G\delta = 0$, $m = 2$), with exact contact orders 2.038 and 4.000 and coefficient agreement to 1.07% and 0.00% with Eq. (4). A mixed direction crosses over near the predicted $\varepsilon_{\mathrm{cross}}$. Insets: Monte Carlo single-look d' exponents 0.95 and 2.05; CUSUM delay slopes -2.16 and -3.92 (two-point confirmations of the densely sampled exact-KL exponents, not independent fits); and the nuisance quotient, which is zero under a free amplitude gauge ($J_\theta = 0$) and restored by a finite anchor ($J_\theta > 0$). All sources are the committed JET_TEST artifact.

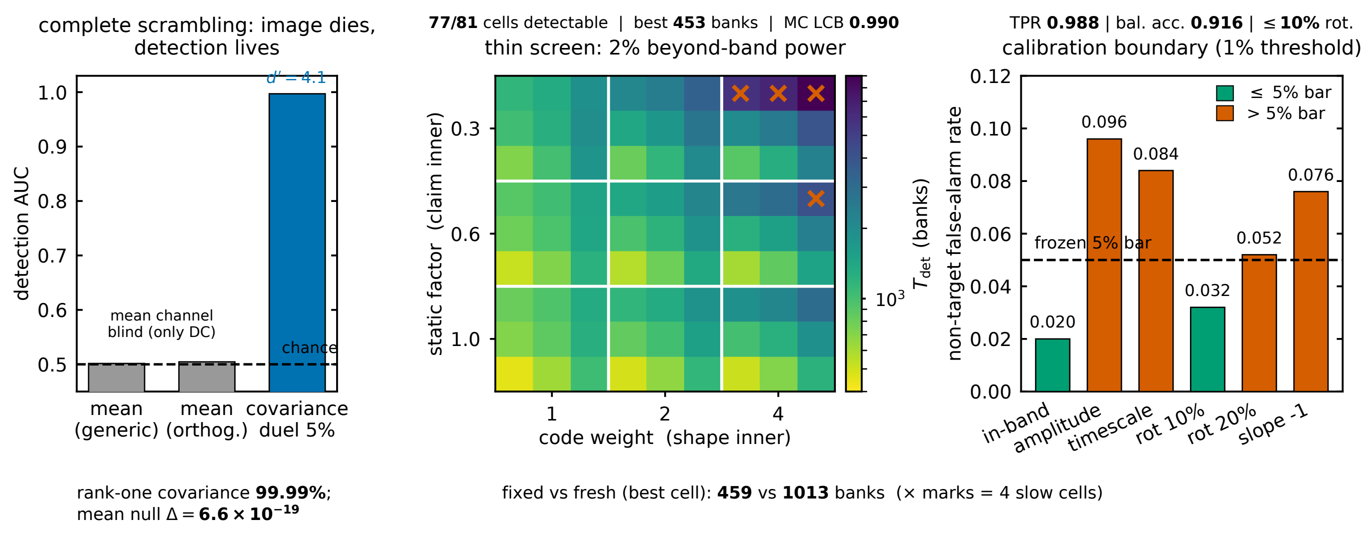

FIG. 3. **Rank collapses, detection survives: two optical endpoints and the honest boundary.** *Left (complete scrambling):* the image channel dies— the mean channel is at chance—while the detection channel lives: the covariance duel reaches $d' = 4.1$, AUC $= 0.997$ for a generic 5% change, with rank-one covariance (99.99%) and mean null 6.6×10^{-19}. *Center (thin-screen power):* the simulated sealed thin-screen detector meets its 2% beyond-band power endpoint across the predeclared operating map (× marks the four slow cells), certifying practical power. *Right (calibration boundary):* at the frozen 1% alarm threshold four-class balanced accuracy is 0.916 and in-band false alarms are 0.020, but medium-amplitude and medium-timescale events give false-alarm rates of 0.096 and 0.084, with robustness frozen to the $\leq 10\%$ aperture-rotation domain—these are the measured attribution limits, not fine print.

End Matter

APPENDIX A: PROOF OF RANK–JET SEPARATION AND THE QUADRATIC SPECIALIZATION

Let $P_x = P_{q(x),\eta}$ with $q : \mathbb{R}^N \to \mathbb{R}^r$ smooth and every nuisance η profiled. By the chain rule the efficient scene score is $Dq_x^\top s_q$, where s_q is the efficient score in functional space, so the scene Fisher matrix is $I_x = Dq_x^\top J_q\, Dq_x$ with $J_q = \mathbb{E}[s_q s_q^\top]$. Since $J_q \in \mathbb{R}^{r\times r}$, $\mathrm{rank}(I_x) \le \mathrm{rank}(Dq_x) \le r$, which is Eq. (2). Rank is therefore controlled by the number of scene functionals the law exposes and is independent of sample size.

For the contact order, expand the profiled Chernoff distance along δ. Write $q(x+\varepsilon\delta)-q(x) = \varepsilon\, u(\delta)+\frac{\varepsilon^2}{2}v(\delta)+o(\varepsilon^2)$ with u, v projected off the nuisance tangent. For a regular reduced law $C_*(\varepsilon;\delta)$ is analytic in the functional displacement and vanishes to second order in it, so if $u \neq 0$ then $C_* = c_1\varepsilon^2 + o(\varepsilon^2)$ and $m = 1$; if $u = 0$ but $v \neq 0$ then $C_* = c_2\varepsilon^4 + o(\varepsilon^4)$ and $m = 2$. Equation (1) then gives $T_{\mathrm{req}} \asymp \varepsilon^{-2m}$.

Specialize to the Gaussian covariance law $\Sigma(x) = \Sigma_0 + z(\varepsilon)H$, $\Sigma_0 \succ 0$, with $K = \Sigma_0^{-1/2} H \Sigma_0^{-1/2}$. The exact divergences are $D(P_\varepsilon \| P_0) = \frac{1}{2}[\mathrm{tr}\, A - \log\det(I + A)] = z^2\|K\|_F^2/4 + O(z^3)$ and, at the symmetric Chernoff point $s = \frac{1}{2}$, $C = z^2\|K\|_F^2/16 + O(z^3)$, with $A = zK$. For the scrambled functional $Q(x) = x^\top Gx$, $G \succ 0$, along $x_\varepsilon = x + \varepsilon\delta$ the covariance shift is $z(\varepsilon) = 2c\,\varepsilon + d\,\varepsilon^2$ with $c = x^\top G\delta$ and $d = \delta^\top G\delta > 0$ (so $d = v/2$ for the curvature $v = 2\delta^\top G\delta$ of the main text). If $c \neq 0$ then $z = 2c\varepsilon + O(\varepsilon^2)$, so $C = c^2\|K\|_F^2\varepsilon^2/4 + o(\varepsilon^2)$ and $m = 1$. If $c = 0$ then $z = d\varepsilon^2$, so $C = d^2\|K\|_F^2\varepsilon^4/16 + o(\varepsilon^4)$ and $m = 2$: the ordinary score vanishes, but the Hessian of Q is $2G \succ 0$, so the law leaves the null at second order in every nonzero first-order-orthogonal direction. Because $d > 0$ for all $\delta \neq 0$, no in-aperture direction is blind, establishing Eq. (4) and the empty blind set. $\square$

APPENDIX B: NUISANCE QUOTIENT, AMPLITUDE-GAUGE COLLAPSE, AND THE MINIMAL ANCHOR

Nuisance profiling is a projection in the Gaussian covariance Hilbert space with inner product $\langle A, B\rangle_0 = \frac{1}{2}\,\mathrm{tr}(\Sigma_0^{-1}A\,\Sigma_0^{-1}B)$. Collecting the nuisance covariance derivatives as columns D_η and writing Π_η for the projector onto their whitened span, the efficient signal direction is $H_\perp = (I - \Pi_\eta)H$ and K is replaced by $K_\perp = \Sigma_0^{-1/2} H_\perp \Sigma_0^{-1/2}$; all contact-order coefficients carry $\|K_\perp\|_F$. If $H_\perp = 0$ the change is unidentifiable from that ledger.

The amplitude gauge is exactly this degeneracy. With an unknown medium amplitude a, $\Sigma = R + a\,Q(x)H$, so $\partial_a\Sigma = QH$ and $\partial_Q\Sigma = aH$ are collinear: zero-lag covariance carries *zero* efficient information for separating the scene functional Q from a. The numerics confirm the collapse to machine zero, both at zero lag and across proportional lags. Restoration requires a signature outside the amplitude tangent: a second grain kernel/wavelength or an independently normalized amplitude anchor. A finite amplitude prior recovered 20% of the naive information, and a second grain kernel recovered a smaller finite fraction—the minimal anchor turns a hidden fatal gauge into a design condition, shown as the amplitude-anchor icon in Fig. 1. This is the exact content of boundary 2 of the theorem.

APPENDIX C: MODEL/PROTOCOL CAPSULE AND INTEGRITY DISCLOSURES

Theorem exams. The jet and scrambling tests use the exact Kronecker speckle generator (fully developed circular-complex-Gaussian field, factorized grain⊗code covariance) on a 32×32 grid, $N = 1024$, code band $P_B = 3$, grain band $K = 13$, $M = 36$ codes, 10^4 photons per bucket, and perform no reconstruction.

Sealed detector. The thin-screen sentinel uses a true zero-dimensional bucket record on a 64×64 scene ($N = 4096$), modulator band $k_p = 5$, $M = 128$ signed effective codes realized by 256 complementary nonnegative exposures per bank at a 20 kHz pattern rate (12.8 ms per bank, cap T_{eff} = 4096 banks = 52.4 s), 10^4 detected photoelectrons per physical bucket, and a quasi-static medium within a bank that is independent between banks. Every engine uses the physical complementary-exposure shot term $|P|x$; the signed-difference mean is never used as photon throughput. The sealed run used 12 confirmatory banks and 2.76 of a 6.0 GPU-hour ceiling. Thresholds were committed before confirmatory generation, and no scene, code, medium realization, anomaly, or seed crosses banks; there was one sealed run and no repair.

Integrity disclosure 1 (record generator). After protocol freeze and before unblinding, the record generator was refactored solely to stream Monte Carlo records in GPU-memory chunks; an audited diff found no changes to hypotheses, thresholds, bank counts, scenes, estimators, or endpoints, although altered random-number consumption produced a different, still-blinded realization set. All reported decisions apply the frozen analysis unchanged to that disclosed chunked stream.

Integrity disclosure 2 (shot model). A mandated independent-engine check exposed a signed-code shot-noise error that had inflated covariance Fisher information by approximately 1.9×; replacing the signed difference Px with the physical complementary-exposure throughput $|P|x$ reduced the frozen $P1$ witness coverage from 0.050 to 0.033 and eliminated the apparent naturalscene living pocket, so all values reported here come from the corrected artifacts and every prior kill only strengthened.

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